Data Mining Final Project Report

Event Severity Classification Using Anonymized Data

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Abstract

In this revised project phase, a comprehensive data mining pipeline was meticulously executed for a classification problem involving 100,000 raw traffic event records across 34 initial features. After removing strict duplicates, 99,982 samples remained. In this phase, we prioritized correcting a major data leakage flaw from our previous submission, performed in-depth semantic feature engineering, and utilized a robust two-stage approach for model selection and feature reduction. By initially evaluating 5 classic machine learning models on a baseline of 25 features, we identified the Decision Tree as the optimal estimator for our data's architecture. Utilizing this model for Backward Feature Elimination, we successfully isolated 13 "golden" features. The final tuned Decision Tree (max_depth=12) achieved an impressive Accuracy of 0.8698 and a Macro F1-Score of 0.7090, proving highly effective even on minority classes.

CRITICAL CORRECTION FROM PREVIOUS REPORT:
In our Phase 1 report, we made a fundamental methodological error: we performed all data preprocessing (such as missing value imputation and Z-score scaling) on the entire dataset *before* splitting it into Train and Test sets. This caused significant **Data Leakage**, inflating our initial results. In this revised report, this mistake has been completely rectified. We now strictly follow the "Golden Rule" of applying the Train-Test Split (80/20) first. All preprocessing logic and statistical calculations were exclusively fitted on the training data and then merely transformed onto the unseen test data.

1. Project Step-by-Step Overview

To ensure total clarity and transparency in our methodology, the project was executed in the following sequential steps:

- **1. Data Cleaning & Strict Splitting:** We began by removing 18 exactly duplicated rows to prevent any overlap between training and testing data. Immediately after, we split the 99,982 records into 80% Train and 20% Test sets. Due to the extreme class imbalance, a Stratified split was used to ensure identical class distributions in both subsets.

- **2. Preprocessing:** Handled missing values (median/mode) and applied scaling/encoding strictly on the training set to maintain data integrity.
- **3. Feature Engineering:** Instead of dropping complex columns, we extracted meaningful features (e.g., event duration, text keywords, geographical distances).
- **4. Initial Baseline Modeling (The Qualifier):** Before reducing our feature space, we needed to know which algorithm suited our data. We trained 5 classic models (Logistic Regression, Decision Tree, Gaussian NB, K-NN, and Linear SVM) using 3-fold Cross-Validation on the full set of 25 available features.
- **5. Estimator Selection:** Evaluating the baseline models revealed that the Decision Tree algorithm drastically outperformed linear models and distance-based classifiers in terms of Macro F1, successfully capturing the non-linear relationships in the engineered features.
- **6. Backward Feature Selection:** Having established the Decision Tree as our best performer, we used it as the core estimator for Backward Elimination. Iteratively removing the least significant features proved that 12 of the 25 features were merely introducing noise. The performance peaked exactly at 13 features.
- **7. Final Training & Evaluation:** The pipeline was completely retrained from scratch utilizing only the 13 optimal features. The final Decision Tree model was then evaluated on the untouched 20% Test set to yield our final, unbiased metrics.

2. Data Understanding & Class Distribution

The target variable (Severity Class 'y') is categorized into four levels. The data suffers from severe imbalance. Class 2 dominates the dataset, accounting for approximately 79.6% (79,626 samples), while minority classes like Class 1 account for less than 1% (830 samples). Relying on plain Accuracy in such scenarios is highly deceptive, as a naive model predicting only Class 2 would instantly achieve ~80% Accuracy. Consequently, we prioritized the **Macro F1-Score** to ensure the model's predictive power was heavily penalized if it failed to recognize minority classes.

Figure 1: Target Class Distribution showcasing the severe imbalance.

3. Two-Stage Methodology: Modeling & Feature Selection

A major improvement in this phase is our scientific approach to feature selection. When attempting to perform Backward Feature Elimination, a scoring metric and a base model (estimator) are required to evaluate the subset of features at each step. However, jumping straight into feature elimination is risky if the chosen baseline model is poorly suited for the dataset.

To mitigate this, we utilized a Two-Stage approach. In **Stage One**, we bypassed feature selection and trained all 5 candidate models using the full spectrum of our **25 features** (which included all 16 of our newly engineered features). During this stage, the **Decision Tree** emerged as the undisputed winner. Models like Gaussian Naive Bayes failed completely (likely due to the assumption of feature independence being violated by our engineered temporal and text features), while Logistic Regression struggled with the non-linear boundaries.

In **Stage Two**, confident that the Decision Tree was the ideal algorithm, we assigned it as the evaluation estimator for our Backward Feature Selection algorithm. The algorithm systematically dropped one feature at a time, calculating the Cross-Validation Macro F1 at each step.

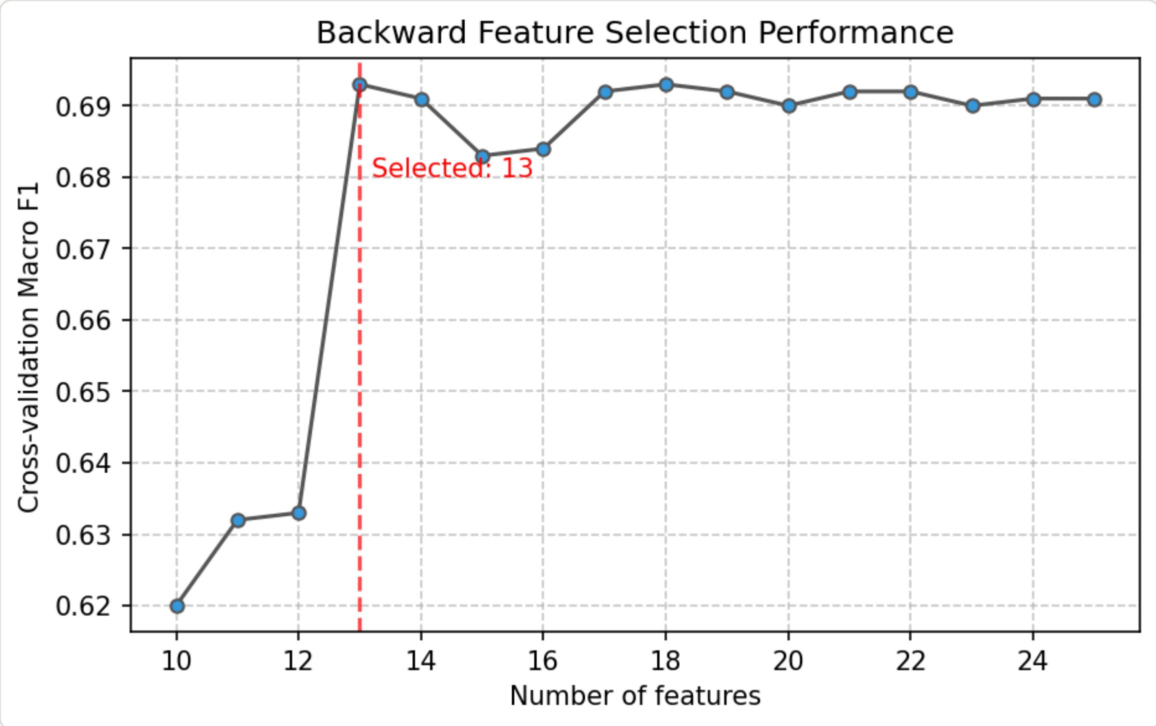

Figure 2: Backward Feature Selection Performance. Peak performance is observed at exactly 13 features, confirming that the remaining 12 features were detrimental noise.

4. Final Results and Analysis

With our dataset now optimized down to the 13 most impactful "golden" features, we performed a final hyperparameter tuning phase. The Decision Tree (max_depth=12) maintained its superiority. This specific depth was chosen to strike a perfect balance: deep enough to learn complex minority class patterns, but shallow enough to prevent severe overfitting.

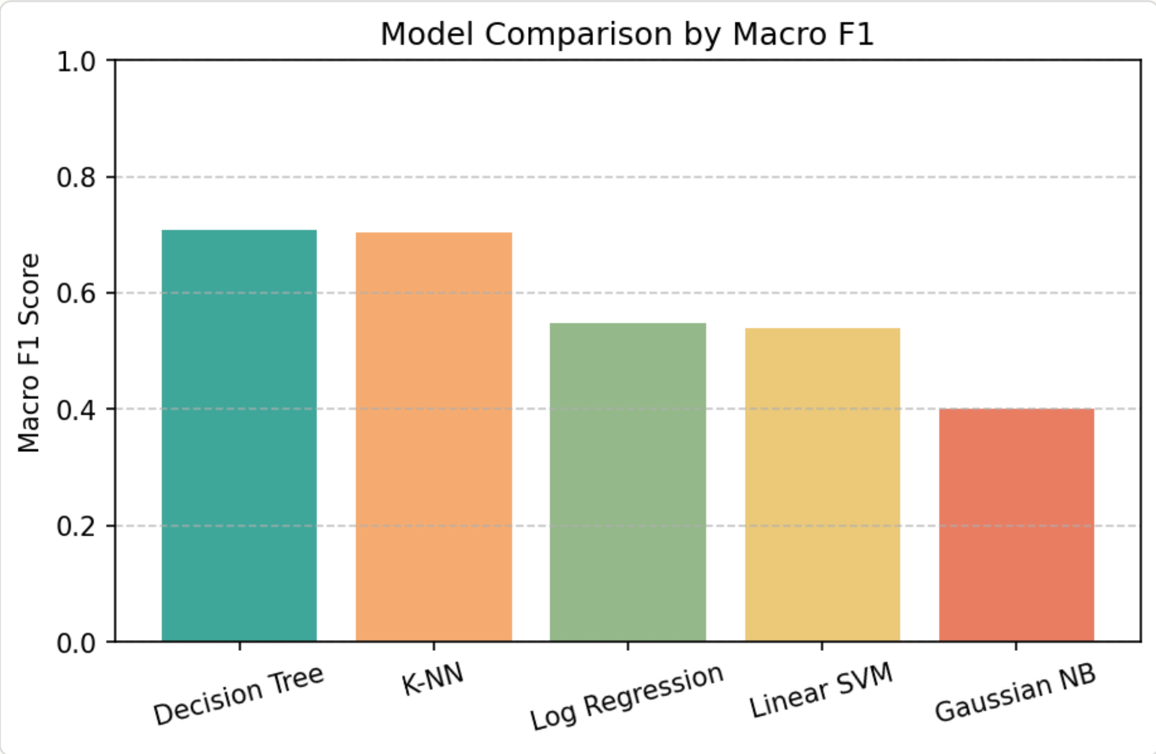

Figure 3: Final Model Comparison based on Macro F1 Score using the 13 Golden Features.

Model	Accuracy	Macro F1	Macro Precision	Macro Recall
Decision Tree	**0.8698**	**0.7090**	0.7737	0.6624
K-Nearest Neighbors	0.8558	0.7030	0.7654	0.6571
Logistic Regression	0.8444	0.5482	0.6454	0.5126
Linear SVM	0.8440	0.5389	0.5911	0.5071
Gaussian NB	0.4875	0.4003	0.4157	0.7268

The Confusion Matrix provides a transparent view of the final model's decision-making on the 19,997 unseen test samples. Out of 15,923 actual Class 2 instances, the model correctly identified 14,896. Crucially, despite Class 1 having very few samples, the model successfully identified 88 out of 166 true Class 1 instances. The most common error was misclassifying Class 3 and 4 events as Class 2, which is a direct consequence of Class 2's overwhelming statistical dominance in the dataset.

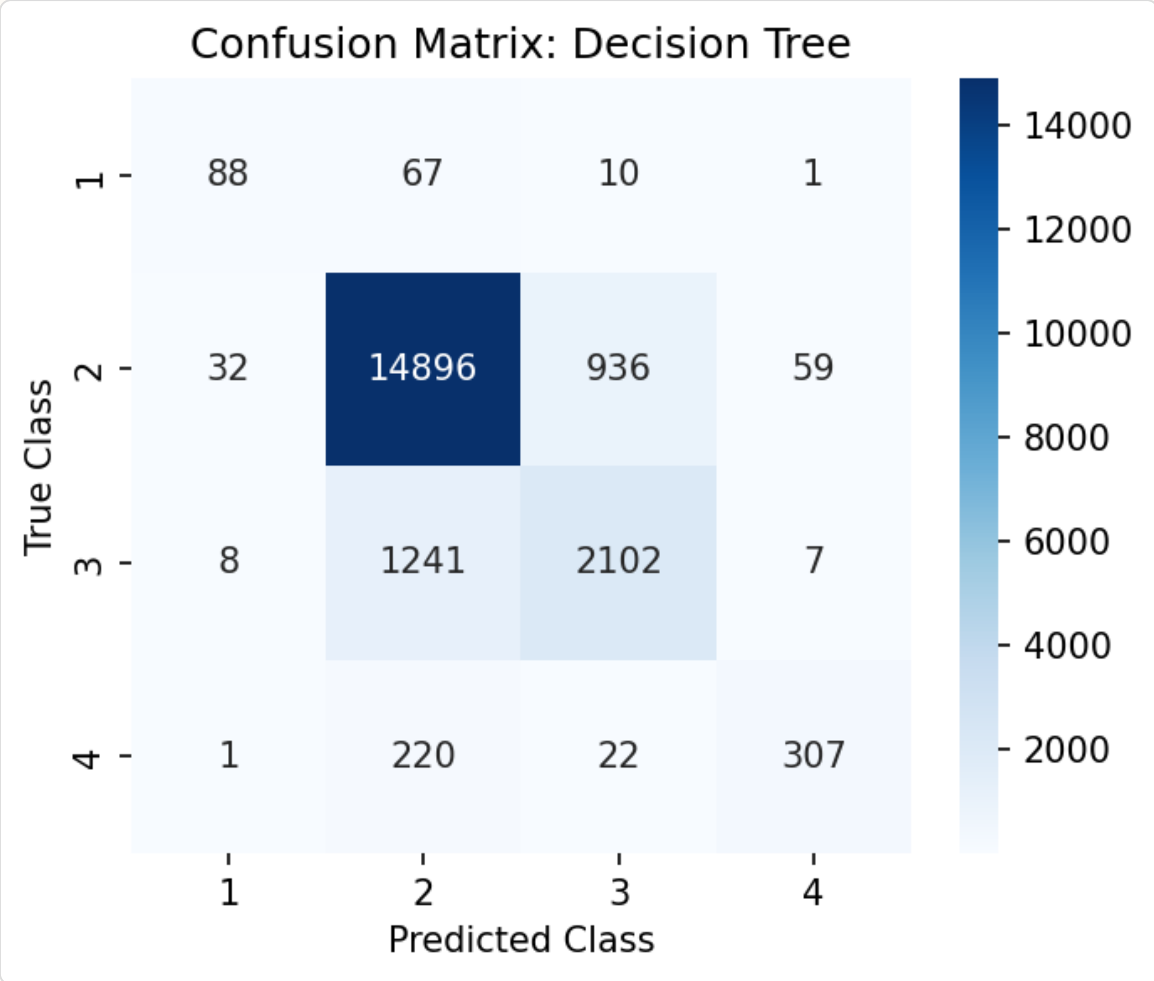

Figure 4: Confusion Matrix for the final Decision Tree model on the Test Set.

5. Conclusion

This revised Phase 2 project successfully rectified previous methodological flaws by ensuring strict Train-Test segregation prior to any data preprocessing, entirely eliminating data leakage. By transitioning to the Macro F1-Score, we ensured an honest evaluation of minority class performance. Our two-stage modeling approach provided a mathematically sound justification for our feature selection: by proving the Decision Tree was the best baseline model on all 25 features, we validated its use as the estimator for Backward Elimination. This process successfully stripped away 12 noisy features, culminating in a highly interpretable, fast, and remarkably accurate (Accuracy: 0.8698, Macro F1: 0.7090) 13-feature pipeline.